

Spring 2001

Problem 1 (Sp01). Let $\mathbb{F}_q$ be the finite field of order q , and let V be a two dimensional vector space over $\mathbb{F}_q$. Find the number of endomorphisms of V that fix at least one nonzero vector.

Problem 2 (Sp83, Sp01, Fa19). Suppose that f is a continuous function on $\mathbb{R}$ which is periodic with period 1. Show that:

1. The function f is bounded above and below and achieves its maximum and minimum.
2. The function f is uniformly continuous on $\mathbb{R}$.
3. There exists a real number x_0 such that $f(x_0 + \pi) = f(x_0)$.

Problem 3 (Sp01). Find all commutative rings R with identity such that R has a unique maximal ideal and such that the group of units of R is trivial.

Problem 4 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21).

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^{2n}}$, where n is a positive integer.

Problem 5 (Sp01). Let α and β be real numbers such that the subgroup Γ of $\mathbb{R}$ generated by α and β is closed. Prove that α and β are linearly dependent over $\mathbb{Q}$.

Problem 6 (Sp01). Let S be a *special orthogonal* $n \times n$ matrix, that is, a real $n \times n$ matrix satisfying $S^T S = I$ and $\det S = 1$.

1. Prove that if n is odd then 1 is an eigenvalue of S .
2. Prove that if n is even then 1 need not be an eigenvalue of S .

Problem 7 (Sp01). Let the functions $f_n : [0, 1] \rightarrow [0, 1]$ for $n = 1, 2, \dots$ satisfy $|f_n(x) - f_n(y)| \leq |x - y|$ whenever $|x - y| \geq 1/n$. Prove that the sequence $\{f_n\}$ has a uniformly convergent subsequence.

Problem 8 (Sp01). If G is a finite group, must $S = \{g^2 \mid g \in G\}$ be a subgroup? Provide a proof or a counterexample.

Problem 9 (Sp01). 1. Prove that an entire function with a positive real part is constant.

2. Prove the analogous result for 2×2 matrix functions: If $F(z) = (f_{jk}(z))$ is a matrix function in the complex plane, each entry f_{jk} being entire, and if $F(z) + F(z)^*$ is positive definite for each z , then F is constant.

Problem 10 (Sp01). Let $M_{n \times n}$ be the vector space of $n \times n$ complex matrices. For A in $M_{n \times n}$ define the linear transformation of T_A on $M_{n \times n}$ by $T_A(X) = AX - XA$. Prove that the rank of T_A is at most $n^2 - n$.

Problem 11 (Sp01). Let U be a nonempty, proper, open subset of $\mathbb{R}^n$. Construct a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that is discontinuous at each point of U and continuous at each point of $\mathbb{R}^n \setminus U$.

Problem 12 (Sp88, Sp01). Let $1 \in S_9$ be the identity permutation. Determine the minimum of all positive integers m such that every $\sigma \in S_9$ satisfies $\sigma^m = 1$. Determine also the minimum of all positive integers m such that every $\sigma \in A_9$ satisfies $\sigma^m = 1$.

Problem 13 (Sp01). Let the power series $\sum_{n=0}^{\infty} c_n z^n$, with positive radius of convergence R , represent the function f in the disc $|z| < R$. For $k = 0, 1, \dots$ let s_k be the k -th partial sum of the series $s_k(z) = \sum_{n=0}^k c_n z^n$. Prove that

$$\sum_{k=0}^{\infty} |f(z) - s_k(z)| < \infty$$

for each z in the disc $|z| < R$.

Problem 14 (Sp01). Consider the *differential-delay equation* $y'(t) = -y(t - t_0)$. Here, the independent variable t is a real variable, the function y is allowed to be complex valued, and t_0 is a positive constant. Prove that if $0 < t_0 < \pi/2$ then every solution of the form $y(t) = e^{\lambda t}$, with λ complex, tends to 0 as $t \rightarrow +\infty$.

Problem 15 (Sp01). Let A be an $n \times n$ matrix over a field $\mathbb{K}$. Prove that

$$\text{rank } A^2 - \text{rank } A^3 \leq \text{rank } A - \text{rank } A^2$$

Problem 16 (Sp01). Let T_0 be the interior of a triangle in $\mathbb{R}^2$ with vertices A, B, C . Let T_1 be the interior of the triangle whose vertices are the midpoints of the sides of T_0 , T_2 the interior of the triangle whose vertices are the midpoints of the sides of T_1 , and so on. Describe the set

$$\bigcap_{n=0}^{\infty} T_n$$

Problem 17 (Sp01). Prove that the polynomial $f(x) = 16x^5 - 125x^4 + 50x^3 - 100x^2 + 75x + 25$ is irreducible over the rationals.

Problem 18 (Sp01). Let f be an entire function such that

$$\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta \leq Ar^{2k} \quad \text{for } 0 < r < \infty$$

where k is a positive integer and A is a positive constant. Prove that f is a constant multiple of the function z^k .